

HR Analytics
Prof. Santosh Rangnekar, Professor
Department of Management Studies
Indian Institute of Technology, Roorkee
Dr. Abhishek Singh, Assistant Professor, OB & HR
Indian Institute of Management, Rohtak
Week: 1
Lecture 05: Developing HR analytics unit and culture

Dear participants, till now we have learned the basic concepts of the data analytics as well as how to manage the data. In this session, we will learn how to develop the HR analytics unit in the organization, what should be the parameter, what we should consider, how we should develop, what are the things that we should think of before developing, during the development, developing the HR unit and how we should promote the analytical culture within the organization, how we should promote the database decision making or data driven decision making within the organization. So, that is what we will learn in today's session. So, let us start with the content of the day. So, this is the content for the day. So, first aspect that is what we will cover that is the need for the HR analytics in the organization.

So, what is the need? What is the need for this HR analytics in the organization or unit? Next thing, who are the stakeholder of this HR analytics? Who are the stakeholder? That is what we will understand. Next, one matrix that is what we will understand that is the complexity and impact matrix. How complex it is? To implement or to develop the HR analytics unit and what are the impacts of that HR analytics unit and then we will discuss how to assess the complexity and how to assess the impact. Right, that is what we will understand.

And third, next, building the capabilities; the moment we have taken a decision, we have to develop the HR analytics unit in our organization, then how we can develop the capabilities, and what we should do. Next, the moment you have decided to develop the capabilities, you need to develop the team that will work on these analytical projects in the organization. So, what should be the role of each team? Team member, right, within the organization and then partner for the skills, right, some of the skills that you may not be having within your organization, for that you have to partner with some other organization, right, and then you have to develop the operating model of your organization for the use of HR analytics. So, this is the content for the day that is what we will discuss in this particular session. So, let us start with one by one.

So, first need for HR analytics in the organization. Most of the organization they have

agreed if we are making the database decision then our decisions are likely to be more accurate as per the need of the organization. So, that gives us the competitive advantage. So, when we are making decisions in an organization, it is all about making good decisions. So, What this analytics says, if we are making the decision on the basis of data, then our decisions are likely to be good.

So, the probability of making good decisions increases. So, when you are making good decisions which result in the development of the business within your organization. So, that will give you the competitive advantage. Second reason, that is what you can see the top management believe. Top management believes that we should make this data-driven decision; they believe in the importance of making the decision through the data.

So, they are, if they believe firmly, yes, we should go ahead and make the decision on the basis of this data within the organization related to the HR to manage the people in the HR. If they are having the strong belief, then they will force to develop the HR analytics unit within the organization. Next reason that could be operational efficiency. So, if we need to increase the operational efficiency of all six functions of the management, all six functions of the HR department. So, all six functions are recruitment, selection, training, development, performance and compensation.

So, for all six functions, if we need to increase operational efficiency and if we are making the decision on the basis of data related to these functions, then our decision operational efficiency is likely to increase among these functions in any organization. So, that could be the one of the factors. Reason why we should develop the HR analytics unit within the organization. So, sometimes, to take a competitive advantage, sometimes because the top management believes in it, sometimes you see a clear impact on operational efficiency. So, the operations that you are doing are related to recruitment, selection, performance, compensation, learning, and development.

So, related to that you can increase the operational efficiency because of that you want to develop the HR analytics unit. So, this operational efficiency may lead to reduced cost, may lead to a reduction in the cost of the operations, and you may save some of the amounts, you can manage the people better, and you can reduce attrition. you can increase the number of applications, you can use the technology, you can use the technology to select the people by framing the benchmark, you can use the method for the training, you can apply this analytics and training so that you can get the better outcome of the training. So, that is how you can reduce the caused by the application of HR analytics directly or indirectly. So, that might be pushing the pressure on the top management.

Yes, we should adopt the HR analytics unit or we should develop the HR analytics unit in our organization. So, these are the some of the reasons why organization wants to develop the HR analytics unit within the organization. So, it could be one of the thing competitive advantage they want to take a competitive advantage they are because of the top management they believe the top management wants to develop and next they see the operational efficiency can be brought through. the recruitment, selection, training, development, performance and compensation and they are putting pressure to reduce the cost. So, these could be the regions to develop the HR analytics unit within the organization.

Next step is when you are thinking to develop the HR analytics unit, these are the regions because of these regions you have decided you want to develop the HR analytics unit. So, my suggestion is always think about the stakeholder, who are the stakeholder. So, business leader, so how this HR analytics is going to serve the business leaders, HR leaders and board of directors. So, specifically I am talking about the top management. If you are going to develop the HR analytics unit within your organization, then how it is going to impact the business leaders, HR leaders and board of directors within the organization, that is what you need to think.

Second category of the stakeholders comes, who are the dependent, the HR analytics is dependent on what? That second category that is what you can see. So, here you can see served, HR analytics is serving these people, business leader, HR leaders and board of directors. And stakeholders who are dependent on this HR, right, HR unit is dependent on stakeholders. So, one stakeholder could be the data owners. Right, you need to take the data from the multiple sources.

So, HR analytics unit is dependent on the data owners. Second one is the technology owners. So, some of the technology that you are using like Tableau, Power BI, right. So, owner of these HR analytics is dependent on them. Third is subject matter expert.

So, these are the three key stakeholders which will ensure that this HR analytics unit will run smoothly within the organization. So, HR analytics unit clearly dependent on these three stakeholders, right. The third type of stakeholder is the employees. What is the category? Impacted. So, first, so how I have divided these stakeholder? One is served.

How this HR analytics unit will serve the stakeholder? To whom it will serve? So, top management, so business leaders, HR leaders, board of directors. Second one is, HR analytics unit is dependent on whom? It is dependent on data owners, technology, and subject matter experts, and the third category of the stakeholder comes the impacted. So, directly impact it has on whom this directly impact because here decision that we are

going to make related to the employee. So, direct impact that you can see that HR analytics unit will have that is on employees because all HR department you, why the data department is there in any organization to manage the HR, employees of the organization. So when the decisions are taken based on the data, so these decisions will impact to whom? They will impact the employees within the organization.

So these are the three stakeholders of the HR, three types of the stakeholders of HR analytics unit. So, whenever you are planning to develop the HR analytics unit within your organization, please think about these stakeholders, how it is going to serve, how it is going to be dependent, to what extent it is going to be dependent and how much impact it is going to have on the employees. Do the assessment of these all stakeholders and then think how it is impacting, how much dependency is there and how it is going to serve. That will help you to develop the mission and vision. Or simply I can say that it can help you to develop the rationale to implement or to develop the HR analytics unit within your organization because without rationale there is no need to develop the HR analytics unit.

There has to be a rationale, there has to be a logic, there has to be importance to develop the HR analytics unit within the organization. So, how you will be able to understand the importance? By analyzing and assessing all stakeholders within the organization. So, I hope you would have understood how to analyze the various types of stakeholder which are associated with the HR analytics unit, right, within the organization. So, let us move to the next. So, next thing that I can ask you to assess your organization where your organization stands.

So, how you can assess? Two things that you can see on this 2 by 2 matrix. So, you can develop this 2 by 2 matrix to see first is complexity, complexity to implement the HR analytics unit or to develop the HR analytics unit. So, in simple way just I ask you how hard it is to develop the HR analytics unit in your organization, right. Then you see the complexity, how complex it is, how to measure the complexity that is what I will explain to you. So, you can see how to measure the complexity.

So, you can measure the politics, whether such kind of initiatives will be appreciated within the organization or not, right. If people are appreciating, they are acknowledging, supporting, backing, right. So, these things are happening in rationales, logics are appreciated, then you can see that shows the how strong politics is there, right. If people are not appreciating or acknowledging the right kind of logics that you are giving to develop this analytical unit in your department then you can say that high level of politics is there in the organization. Second thing that you can see based on available skills, so in your department whoever is there, like 5 employees are there, 6 employees are there, what is their skill set, right, what kind of skills that currently they are having.

Third is availability of data. The data, how much data is available and what is the authenticity of that data, how pure that data is, how much you can trust to that particular data. So, that is what you can check. And fourth thing that you should have about the technology. So, how much technology is there because you can see that data warehouse that HRIS, Human Resource Information System. So, if your organization is not developed, then you may not be having this developed HRIS, right, it is quite possible.

So, if that HRIS is not developed, then you may face difficulty to even collect the raw data also, right. You may not be able to collect the raw data, it may be very, very difficult. So, if politics is high, skill level is low, availability of data is low and technology is low. When use of technology within the organization is low, then you can say it is complex task, right. If I reverse it, politics is low, skill is high, availability of data is available and use of technology is high, then you can say complexity is low.

In first case, complexity was high because politics was high, skill level was low, data was, availability of data was low and use of technology also very low within the organization. So, then you can say that the, to implement, to develop the HR analytics unit is very difficult. But if we reverse the situation, politics is low, skill is high, data set is high, technology is high, then we can say that the complexity level is low. So, this is the one dimension of this 2 by 2 matrix. So, that is how you can measure the complexity.

Next thing it comes impact. Impact means how beneficial this HR analytics unit would be for your organization. So, how much useful it is? So, how we will measure this? So, for this we are having the three parameter, right. So, first parameter is return on investment. Whatever amount that we are going to invest, whether we will be able to recover that amount or not, right. So if you see the recovery of that amount, then you can say yes, it is having the high impact.

Second that you can say that timing, so whatever time that you can save by applying this HR analytics tools and technique in taking the decisions, right. So that is the timing that you can say and then opportunity cost. So based on these three parameter you can decide the impact of this HR analytics unit within your organization. So the moment you have completed the assessment of impact and complexity of developing the HR analytics unit and then you see on a 2 by 2 matrix in which quadrant you are, whether it is a quick win. It means, quick win means impact is high and complexity is low, right.

So, it is a very good situation for the organization. They should go ahead and implement, or they should develop the HR analytics unit because complexity level is low. It means effort required to develop the HR analytics unit is going to be very less. It is going to be

less because complexity level is low, but impact is very high.

So, it is in favour of the organisation. Second is complexity is low, impact also low. So, just you can put it into the future plan. In future you may think about it. If next situation is complexity is very high and impact also very high.

So, this is the big bet. So, here again you can see whether readiness to take a risk, readiness to take a risk. if organization is ready to take a risk here like impact is high and complexity is high. So, sometime what happens in this situation when complexity is too high. So, entire focus of the department goes in that direction. So, if you feel it is the top most job to develop the HR analytics unit in your organization, that is the first priority, then you can go ahead.

Otherwise, if your department is having some other priorities, so just prioritize the work. So, if some other priorities are there, then you can think about the some other priorities and then you can plan for this particular thing. So, that is what you need to check. But last case that is what you can see here pet project if complexity is very high and impact is very low. So, there is no need to develop in the right situation.

Again, you can put it in the future because the complexity is very high, which means it will require huge resources, but the impact is not that high. So, the impact is not that high but resources are huge. So, resources are huge, so in that case you can think you may postpone it. So, in these two situations when quick win and big bet is there, in these two situations you can develop the HR analytics unit after assessing the complexity level and impact of that HR analytics unit within the organization. So, that is how you can assess the situation of your organization within your organization, and then you can make a decision on whether you should develop the HR analytics unit now or after some time.

So, building the capabilities. The moment you have taken a decision, you have to develop the HR analytics unit within the organization right. So, now you have to build the capabilities of your unit. So, how you can develop? So, few things that you need to remember. First thing that you need to remember that is the know your data. So, whatever data sources are there you need to see where the data is available in the form of text or it is available in the form of net number.

There are only two types of data generally that you will see either in the form of text or in the form of numbers. So, the text could be one paragraph, right, appreciation somebody has written an appreciation for one employee. So, that is the textual data. Number that is what you can say that age, performance, rating, right. So, basically you can understand the numbers 1, 2, 3, 4, 5, 6, 7, right.

So, whether data is available in that form or it is available in the form of text, whether it is structured data or unstructured data like data is ready for analysis or not ready. If data is unstructured, then you have to transform that particular data. So, how much time it will take, do you have capabilities or not, that is what you need to think. So, first you need to understand what type of data that you are going to use. Know your technology, so the use of technology that HRIS, statistical methods, and visualization technology.

So, know about your technology, availability of your technology. Build the analytical team. After So, these are the data, technology and then team. That is what you need to, building a team, so what you need to understand that for successful team, you need to have a six characteristics.

So, one is business acumen. So, whoever is working in the analytical team, that person should, that team member should have the financial literacy political attuteness, internal awareness and external awareness. So, basically, I would say that there are basic basics of business that a person should understand because if that person is not able to understand the business, then he will not be able to link the HR unit decision with the organization-level decisions, right? So, that, that understanding is must for a team. So, the first thing that, that is a second aspect is that the team must have the knowledge and skill of consulting. So, I learned how to develop problem definition, hypothesis development, project management, solution development, change management, and stakeholder management so that consulting skills are within the organization correctly.

Next building the capabilities continued. So, human resource the knowledge of this human resource functions like that should be there and basic ethical standard people should follow because that HR data is critical, right. So, lot of information is private involved, information is available related to the employees. So, that ethical standard should be there when you will assess that data, analyse this data. So, you may get a lot of information that you should not reveal to others, so that people will have a basic level of ethics that should be there within the team member.

Next one is the work psychology. Like so people should have the basic knowledge about the organizational psychology, right? So leadership, motivation, learning pattern, right? So these are the things they should be able to understand. And then basics of this research design and research methodology they should understand. Next is the basic concept related to the data science like data awareness, qualitative and quantitative methods. And next one is the communication.

So, these are the six skills that a team should have. I am not saying one individual should

have these all skills, one team should have good HR analytical team should have these 6 skills. So, what you have to do? The moment you have identified these 6 skill, now you need to hire the people in such a way so that they are having these 6 skill set. So, now you need to identify first what type of project has to be delivered, what are the skills that are required among these six skills, how many people you have, what are the skills they have, and now what kind of people that you need to hire. So, after understanding these six skills, you need to check with your team about skills that are already available and which skills you need to hire. So, the moment you have decided to hire the people, then you can hire the people, and you can give the responsibilities whatever skill that is not that was not available within the team.

So, that is how what you have to do. You have to identify the team members, and to each team member, you have to give some specific responsibility which will fulfill one of the skills, right? One of the skills that you can assess and then you can give a responsibility to the each team member, right? So, that is how you can develop a team related to it. After doing the recruitment, you also find some of the skills which you can. For some of the skills, you cannot hire the people that those skills are very rare, and you did not find the people within the organization right. Then what you can do? You can partner with some of the organization who have those skills like technology related, or somebody who has excellent knowledge of Excel, IT, right, for such kind of things if you believe your organization is not having the skill, then you can partner with some of the organization and then you can outsource the work related to those skills.

That is what you can do. So, that is how you can design and finally you have to develop one operating model. So, after developing the team you need to develop a operating model. So, you need to link this business HR analytics unit with the strategy of the organization. You need to develop the governance and how it is going to work. And then you have to implement, and finally, you have to fix the responsibility for each task.

So, I hope that is how you would have learned how to develop the HR analytics unit and culture within the organization. Thank you.